

1. Logic, Sets and Numbers

1.1 Logic

Def. 1. A **mathematical Statement** is a well-formulated assertion that is strictly either true or false.

Def. 2. A **Statement** that depends on one or multiple variables is called a **predicate** and denoted by $P(x), Q(x), \dots$.

Def. 3. **Quantifiers** are short hand notation for making statements about predicates.

- $\forall x \in \mathbb{R} : P(x) \iff$ for all $x \in \mathbb{R}$ holds $P(x)$
- $\exists x \in \mathbb{R} : P(x) \iff$ there exists $x \in \mathbb{R}$ s.t. $P(x)$
- $\exists! x \in \mathbb{R} : P(x) \iff$ there exists exactly one $x \in \mathbb{R}$ s.t. $P(x)$
- $\nexists x \in \mathbb{R} : P(x) \iff$ there exists no $x \in \mathbb{R}$ s.t. $P(x)$

Imp. 4. The order of quantifiers matters. $\forall x \exists y$ is not equivalent to $\exists y \forall x$.

Def. 5. Basic **logical connectives** are defined as:

- $A \wedge B :=$ A and B are true
- $A \vee B :=$ A or B is true
- $A \implies B :=$ A implies B is true
- $A \iff B :=$ A is equivalent to B is true

Lem. 6.

$$\neg A \vee B \iff A \implies B$$

Thm. 7. The following equivalences hold:

$$\neg(\forall x P(x)) \iff \exists x \neg P(x)$$
$$\neg(\exists x P(x)) \iff \forall x \neg P(x)$$

aswell as

$$\forall x (P(x) \wedge Q(x)) \iff (\forall x P(x)) \wedge (\forall x Q(x))$$
$$\exists x (P(x) \vee Q(x)) \iff (\exists x P(x)) \vee (\exists x Q(x))$$

1.2 Sets

Def. 8. A **set** is a collection of elements, that are listed or characterized by a certain property. The empty set $:= \emptyset$ and is the only set that contains no elements.

Def. 9. Let A and B be sets. A is a **subset** of B denoted by $A \subset B$ iff.

$$\forall x (x \in A \implies x \in B)$$

A is a **proper subset** of B denoted by $A \subsetneq B$ iff.

$$A \subset B \wedge A \neq B$$

Def. 10. Let A and B be sets. The **union** of A and $B := A \cup B := \{x \mid x \in A \vee x \in B\}$.

Def. 11. The **intersection** of A and $B := A \cap B := \{x \mid x \in A \wedge x \in B\}$.

Def. 12. The **difference** of A and $B := A \setminus B := \{x \mid x \in A \wedge x \notin B\}$.

Def. 13. The **Cartesian product** of A and $B := A \times B := \{(a, b) \mid a \in A \wedge b \in B\}$.

Rem. 14. Note that if $B \subset A$ then the diffrence $A \setminus B$ is called the complement of B in A and $:= B^c$.

1.3 Numbers

Rem. 15. Some important subsets of the real numbers are

$$\mathbb{R}_0^+ / \mathbb{R}^+$$

$$\mathbb{R}_0^- / \mathbb{R}^-$$

which define the intervalls

$$[0, \infty) / (0, \infty)$$

$$(-\infty, 0] / (-\infty, 0)$$

Def. 16. Continuous subsets are called **intervalls**.

Def. 17. An intervall is called **open** if it does not contain its endpoints

$$I = (a, b) := \{x \mid a < x < b\}$$

closed if it contains its endpoints

$$I = [a, b] := \{x \mid a \leq x \leq b\}$$

half-open if it contains one endpoint and not the other

$$I = [a, b) := \{x \mid a \leq x < b\}$$

Rem. 18. An arbitrary subset of $\mathbb{R}$ is called open if it is a union of open intervalls and closed if its complement is open.

Def. 19. An **upper (lower) bound** of a subset $X \subset \mathbb{R}$ is an element $y \in \mathbb{R}$ such that:

$$\forall x \in X : x \leq (\geq) y$$

The **maximum (minimum)** of a subset $X \subset \mathbb{R}$ is the element $x_0 \in X$ such that:

$$\forall x \in X : x \leq (\geq) x_0$$

Often denoted as **max(X)** and **min(X)**.

Def. 20. An intervall $I = (a, b)$ is **bounded** if there exist upper and lower bounds hence

$$a > -\infty \quad \text{and} \quad b < \infty$$

Rem. 21. A bounded and closed intervall is often refered to as **compact**.

Def. 22. The **Supremum** $\sup(X)$ of a subset $X \subset \mathbb{R}$ is the least upper bound of X . and can be characterized by:

$$\forall x \in X : x \leq \sup(X) \wedge$$

$$\forall \epsilon > 0 : \exists x \in X : x > \sup(X) - \epsilon$$

Def. 23. The **Infimum** $\inf(X)$ of a subset $X \subset \mathbb{R}$ is the greatest lower bound of X . and can be characterized by:

$$\forall x \in X : x \geq \inf(X) \wedge$$

$$\forall \epsilon > 0 : \exists x \in X : x < \inf(X) + \epsilon$$

Thm. 24. The **maximum and minimum are unique** aslong as they exist.

Def. 25. **Order Axioms** reflexivity:

$$\forall x \in X : x \leq x$$

antisymmetry:

$$\forall x, y \in X : x \leq y \wedge y \leq x \implies x = y$$

transitivity:

$$\forall x, y, z \in X : x \leq y \wedge y \leq z \implies x \leq z$$

totality:

$$\forall x, y \in X : x \leq y \vee y \leq x$$

Def. 26. A set X is said to be **(order)complete** iff for every non empty subsets $A, B \subseteq X$

$$\forall a \in A \forall b \in B : a \leq b$$

$$\implies \exists c \in X \quad \text{s.t.}$$

$$\forall a \in A : a \leq c \wedge$$

$$\forall b \in B : c \leq b$$

Ex. 27. The real numbers $\mathbb{R}$ satisfy axiomatic properties for addition, multiplication, distributivity and order. Furthermore $\mathbb{R}$ is an order complete field.

Ex. 28. $\mathbb{Q}$ is not order complete. Let $A = \{x \in \mathbb{Q} \mid x^2 \leq 2\}$ and $B = \{x \in \mathbb{Q} \mid x^2 \geq 2\}$. Then $\forall a \in A \forall b \in B : a \leq b$ but there is no $c \in \mathbb{Q}$ such that $c^2 = 2$.

Thm. 29. Every **non empty, bounded from above (below) subset** $X \subset \mathbb{R}$ has a **supremum (infimum)**.

Thm. 30. The **Archimedian Principle** says that For $x \in \mathbb{R}$ and $y > 0$ there exists an element

$$n \in \mathbb{N} \quad \text{s.t.} \quad n \cdot y > x$$

For every $\epsilon > 0$ there exists an element $n \in \mathbb{N}$ such that

$$\frac{1}{n} < \epsilon$$

Thm. 31. The Young Inequality states that for all $\epsilon > 0$ and all $x, y \in \mathbb{R}$:

$$2|xy| \leq \epsilon x^2 + \frac{1}{\epsilon} y^2$$

1.3.1 Vectors and Vector Spaces

Def. 32. The **euclidian space** $\mathbb{R}^n$ is the set of all vectors $x = (x_1, x_2, \dots, x_n)$ where $x_i \in \mathbb{R}$.

Def. 33. **Addition and scalar multiplication** are defined as:

$$x + y = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$$

$$\lambda x = (\lambda x_1, \lambda x_2, \dots, \lambda x_n)$$

Def. 34. A **vector space** has to satisfy the following axioms: Commutativity:

$$x + y = y + x$$

Associativity:

$$x + (y + z) = (x + y) + z$$

Identity element:

$$\exists e \in X \quad \text{s.t. } x + e = x \quad \forall x \in X$$

Inverse element:

$$\forall x \in X \exists x^{-1} \in X \quad \text{s.t. } x + x^{-1} = e$$

Distributivity:

$$x \cdot (y + z) = x \cdot y + x \cdot z$$

Def. 35. For $x, y \in \mathbb{R}^n$, the **standard scalar product** is $:=$:

$$x \cdot y = \langle x, y \rangle := \sum_{i=1}^n x_i y_i$$

The **euclidean norm** is:

$$\|x\| := \sqrt{\sum_{i=1}^n x_i^2}$$

The **euclidean distance** between x and y is:

$$d(x, y) := \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Thm. 36. The triangle- and the cauch-schwarz inequality hold for all $x, y, z \in \mathbb{R}^n$:

$$\|x - z\| \leq \|x - y\| + \|y - z\|$$

$$|\langle x, y \rangle| \leq \|x\| \|y\|$$

1.3.2 Complex Numbers

Def. 37. The **complex numbers** $\mathbb{C}$ are the set of numbers

$$\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}\}$$

where a is the **real part** $\text{Re}(z)$, and b is the **imaginary part** $\text{Im}(z)$.

Def. 38. Arithmetic of complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$: Addition:

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

Multiplication:

$$z_1 \cdot z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2)$$

Def. 39. The **complex conjugate** of $z = x + iy$ is $\bar{z} = x - iy$.

Imp. 40. Division is calculated by expanding with the conjugate of the denominator:

$$\frac{z}{w} = \frac{z \cdot \bar{w}}{w \cdot \bar{w}} = \frac{z \cdot \bar{w}}{|w|^2}$$

Lem. 41. Some usfull properties of the conjugate:

$$z \cdot \bar{z} = |z|^2$$

$$z + \bar{z} = 2\text{Re}(z)$$

$$z - \bar{z} = 2i\text{Im}(z)$$

$$\overline{z + w} = \bar{z} + \bar{w}$$

$$\overline{z \cdot w} = \bar{z} \cdot \bar{w}$$

Def. 42. Geometrically, complex numbers can be represented in the Gaussian plane. The **absolute value** of $z = a + ib$ is its **distance from the origin**:

$$|z| = \sqrt{a^2 + b^2}$$

The distance between z_1 and z_2 is $d = |z_1 - z_2|$.

Thm. 43. The Euler's formula states that for $t \in \mathbb{R}$:

$$e^{it} = \cos(t) + i \sin(t)$$

This allows writing trigonometric functions using complex exponentials:

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}$$

$$\sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

Rem. 44. The **Intuition** behind eulers formula is we can use Taylor series to show that:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Using $x = it$ we get:

$$e^{it} = \sum_{n=0}^{\infty} \frac{(it)^n}{n!} = \dots = \cos(t) + i \sin(t)$$

Def. 45. For $z \in \mathbb{C}$, the polar-coordinates are defined as:

$$z = r(\cos(t) + i \sin(t)) = r e^{it}$$

where $r = |z|$ is the absolute value and $t = \arg(z)$ is the argument.

Rem. 46. To convert from cartesian to polar coordinates:

$$r = \sqrt{a^2 + b^2}$$

$$t = \arctan\left(\frac{b}{a}\right)$$

Note that the arctan function only returns values in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so we have to adjust the angle based on the quadrant of the complex number.

Ex. 47. Multiplication and division in polar-coordinates is much easier than in cartesian coordinates:

$$z_1 \cdot z_2 = r_1 e^{it_1} \cdot r_2 e^{it_2} = r_1 r_2 e^{i(t_1 + t_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1 e^{it_1}}{r_2 e^{it_2}} = \frac{r_1}{r_2} e^{i(t_1 - t_2)}$$

Ex. 48. So is exponentiation or taking roots:

$$z^n = (r e^{it})^n = r^n e^{int}$$

$$\sqrt[n]{z} = \sqrt[n]{r} e^{i \frac{t + 2\pi k}{n}}, \quad k \in \{0, 1, \dots, n-1\}$$

Def. 49. The n -th roots of unity are the solutions to $z^n = 1$, which are given by:

$$z_k = e^{i \frac{2\pi k}{n}}, \quad k \in \{0, 1, \dots, n-1\}$$

Thm. 50. The sum of the roots of unit is always zero.

$$\sum_{k=0}^{n-1} e^{i \frac{2\pi k}{n}} = 0$$

Thm. 51. The **fundamental theorem of algebra** states that every polynomial of degree $n \geq 1$ has exactly n roots in $\mathbb{C}$ (counting multiplicity). These roots appear in konplex-conjugate pairs.

2. Sequences and Series

2.1 Sequences

Def. 52. A **sequence** (Folge) $(a_n)_{n \in \mathbb{N}_0}$ is an infinite list of numbers, mathematically a mapping $f: \mathbb{N}_0 \rightarrow X$. It can be given explicitly or recursively as

$$a_n = f(n) \quad \text{or} \quad a_n = g(a_{n-1}, a_{n-2}, \dots)$$

Def. 53. A sequence (a_n) is called **convergent** with limit $L \in \mathbb{R}$ if:

$$\forall \epsilon > 0 \exists N > 0 \quad \text{s.t.}$$

$$\forall n > N : |a_n - L| < \epsilon$$

Written as $\lim_{n \rightarrow \infty} a_n = L$. If no such L exists, the sequence is **divergent**.

Rem. 54. A special case of divergent sequences are sequences which diverge to $\pm\infty$. Their limits are called **improper limits**.

Thm. 55. The limit of a sequence is unique aslong as it exists.

Thm. 56. Any subsequence $(a_{n_k})_{k \in \mathbb{N}_0}$ of a convergent sequence converges to the same limit L .

Def. 57. An **accumulation point** $A \in \mathbb{R}$ of a sequence fulfills:

$$\forall \epsilon > 0 \forall N \in \mathbb{N}_0 \exists n \geq N$$

$$|a_n - A| < \epsilon$$

Thm. 58. A is an accumulation point $\iff$ there exists a convergent subsequence with limit A .

Thm. 59. Every convergent sequence has exactly one accumulation point (its limit).

Def. 60. A sequence is **upper (lower) bounded** if there exists M such that $|a_n| \leq (\geq) M \quad \forall n \in \mathbb{N}_0$.

Thm. 61. Every convergent sequence is bounded.

Thm. 62. Bolzano-Weierstrass Theorem: Every bounded sequence has an accumulation point.

Thm. 63. Limit arithmetic: For $\lim a_n = K$, $\lim b_n = L$ and $C \in \mathbb{R}$:

$$\lim(a_n \pm b_n) = K \pm L$$

$$\lim(C \cdot a_n \cdot b_n) = C \cdot K \cdot L$$

$$\lim(a_n/b_n) = K/L$$

Rem. 64. Important limits:

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$$

$$\lim_{n \rightarrow \infty} n^{1/n} = 1$$

$$\lim_{n \rightarrow \infty} x^{1/n} = 1 \quad (x > 0)$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$$

$$\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$$

Thm. 65. Sandwich-Theorem: If $a_n \leq b_n \leq c_n$ for $n \geq N_0$, and $\lim a_n = \lim c_n = L$, then $\lim b_n = L$.

Def. 66. A sequence is **monotonically increasing (decreasing)** if $a_n \leq (\geq) a_m$ for $m > n$.

Thm. 67. Monotone Convergence Theorem: Every bounded monotonic sequence converges to its supremum (infimum).

Def. 68. The limit of the supremum is called **limes superior** and denoted as:

$$\limsup a_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} a_k$$

The limit of the infimum is called **limes inferior** and denoted as:

$$\liminf a_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} a_k$$

Def. 69. A sequence is a **Cauchy sequence** iff

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that}$$

$$\forall m, n \geq N : |a_n - a_m| < \epsilon$$

Thm. 70. Every Cauchy sequence is bounded.

Thm. 71. A sequence of real numbers converges iff it is a Cauchy sequence.

2.2 Series

Def. 72. A **series** is an infinite sum $\sum_{i=0}^{\infty} a_i$ of the terms of a sequence (a_n) .

Def. 73. A series is said to converge iff the sequence of partial sums converges.

$$\sum_{i=0}^{\infty} a_i = \lim_{n \rightarrow \infty} \sum_{i=0}^n a_i$$

Rem. 74. Some examples of series are: **geometric series**:

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

for $|x| < 1$ **harmonic series**:

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

diverges for $s > 1$

Thm. 75. Necessary condition for convergence: for $\sum_{i=0}^{\infty} a_i$ to converge we need $\lim_{n \rightarrow \infty} a_n = 0$ i.e. it is a null sequence.

Thm. 76. For **series with non-negative terms** $a_n \geq 0$, the sequence of partial sums is monotonically increasing and **converges iff it is bounded**.

Def. 77. A series is **absolutely convergent** if $\sum |a_n|$ converges. A **conditionally convergent** series converges, but not absolutely.

Thm. 78. Riemann's rearrangement theorem: Let $\sum a_n$ be a conditionally convergent series. Then for any $x \in \mathbb{R}$, there exists a rearrangement of the series that converges to x .

Thm. 79. Comparison test (Vergleichskriterium): If $c_n \leq b_n \leq a_n$ for $n \geq N$: If $\sum a_n$ converges, then $\sum b_n$ converges (**Majorantenkriterium**). If $\sum c_n$ diverges, then $\sum b_n$ diverges (**Minorantenkriterium**).

Thm. 80. Leibniz Criterion: For an alternating series $\sum (-1)^n a_n$ with a monotonically decreasing null sequence (a_n) , the series converges.

Thm. 81. Cauchy Criterion: The series $\sum a_n$ converges iff

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that}$$

$$\forall m, n \geq N : \left| \sum_{k=n}^m a_k \right| < \epsilon$$

Thm. 82. Root Criterion (Wurzelkriterium): Let $\rho = \limsup \sqrt[n]{|a_n|}$. If $\rho < 1$, abs conv; if $\rho > 1$, div.

Thm. 83. Ratio Criterion (Quotientenkriterium): Let $\rho = \lim \left| \frac{a_{n+1}}{a_n} \right|$. If $\rho < 1$, abs conv; if $\rho > 1$, div.

Def. 84. A **power series** (Potenzreihe) centered at a has the form

$$\sum_{k=0}^{\infty} c_k (x-a)^k$$

Thm. 85. Every power series has a **convergence radius** R , with the **interval of convergence** $(a-R, a+R)$. R is computed by the formula $R = \frac{1}{\limsup |c_n|^{1/n}}$.

3. Functions

Def. 86. A **function** $f : D \rightarrow C$ is a map from a domain D to a codomain C s.t. for each $x \in D$ there is exactly one $y \in C$ st. $f(x) = y$ (well defined).

Def. 87. A function f is called **bounded** if there exists $M \in \mathbb{R}$ such that $|f(x)| \leq (\geq) M$ for all $x \in D$.

Def. 88. A function f is called **injective** iff

$$\forall x_1, x_2 \in A : f(x_1) = f(x_2) \implies x_1 = x_2$$

surjective iff

$$\forall y \in B \exists x \in A : f(x) = y$$

and bijective iff f is injective and surjective.

Def. 89. The **restriction** of f under $D' \subset D$:=

$$f \upharpoonright_{D'} : D' \rightarrow C, x \mapsto f(x) \quad \forall x \in D$$

Def. 90. Let $f : D \rightarrow C$ and $g : C \rightarrow B$ be functions then

$$g \circ f : D \rightarrow B \quad \text{with}$$

$$g \circ f(x) = g(f(x)) \quad \forall x \in D$$

is called the **composition** of f and g .

Def. 91. The **identity function** $id_D : D \rightarrow D := id_D(x) = x \quad \forall x \in D$.

Def. 92. If a function $f : D \rightarrow C$ is bijective then there exists a unique function $f^{-1} : C \rightarrow D$ such that

$$f(f^{-1}(y)) = id_C = y \quad \forall y \in C$$

and

$$f^{-1}(f(x)) = id_D = x \quad \forall x \in D$$

f^{-1} is called the **inverse function** of f .

Def. 93. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called **even** iff

$$\forall x \in \mathbb{R} : f(-x) = f(x)$$

and **odd** iff

$$\forall x \in \mathbb{R} : f(-x) = -f(x)$$

Def. 94. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called (**strictly**) **monotonically increasing** iff

$$\forall x_1, x_2 \in \mathbb{R} : x_1 < x_2 \implies f(x_1) < f(x_2)$$

and (**strictly**) **monotonically decreasing** iff

$$\forall x_1, x_2 \in \mathbb{R} : x_1 < x_2 \implies f(x_1) > f(x_2)$$

Thm. 95. Every strictly monotonic function is injective.

3.1 Limits in the context of functions

Def. 96. Let $f : \mathbb{D} \rightarrow \mathbb{R}$ be a function satisfying $\mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \neq \emptyset$ for all $\delta > 0$ and $x_0 \in \mathbb{R}$. We say that $\lim_{x \rightarrow x_0} f(x) = L$ iff

$$\forall \epsilon > 0 \exists \delta > 0 \text{ s.t.}$$

$$x \in \mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \implies |f(x) - L| < \epsilon$$

The limit L is unique.

Def. 97. A function $f : \mathbb{D} \rightarrow \mathbb{R}$ has in x_0 a **right-sided limit** $\lim_{x \searrow x_0} f(x) = L$ iff $\lim_{x \rightarrow x_0^+} f(x) = L$ iff

$$\forall \varepsilon > 0 \exists \delta > 0 \quad \text{s.t.}$$

$$\forall x \in \mathbb{D}(x_0 < x < x_0 + \delta \implies |f(x) - L| < \varepsilon)$$

Def. 98. A function $f : \mathbb{D} \rightarrow \mathbb{R}$ has a **limit for x approaching infinity** $\lim_{x \rightarrow \infty} f(x) = L$ iff

$$\forall \varepsilon > 0 \exists M > 0 \quad \text{s.t.}$$

$$\forall x \in \mathbb{D}(x > M \implies |f(x) - L| < \varepsilon)$$

Def. 99. A function $f : \mathbb{D} \rightarrow \mathbb{R}$ has in x_0 a **improper limit approaching infinity** $\lim_{x \rightarrow x_0} f(x) = \infty$ iff

$$\forall N > 0 \exists \delta > 0 \quad \text{s.t.}$$

$$\text{for all } x \in \mathbb{D}(0 < |x - x_0| < \delta \implies f(x) > N)$$

Thm. 100. Let $\lim_{x \rightarrow x_0} f(x) = K (\neq \pm\infty)$ and $\lim_{x \rightarrow x_0} g(x) = L (\neq \pm\infty)$

$$\lim_{x \rightarrow x_0} (f(x) + g(x)) = K + L$$

$$\lim_{x \rightarrow x_0} (f(x) - g(x)) = K - L$$

$$\lim_{x \rightarrow x_0} (f(x)g(x)) = KL$$

$$\lim_{x \rightarrow x_0} (f(x)/g(x)) = K/L$$

$$\lim_{x \rightarrow x_0} (Af(x)) = AK$$

$$\lim_{x \rightarrow x_0} (f(x)/g(x)) = K/\text{Lif } L \neq 0$$

3.2 Continuity of functions

Def. 101. A function $f : D \rightarrow \mathbb{R}$ is called **continuous** at x_0 if

$$\forall \epsilon > 0 \exists \delta > 0 \text{ such that}$$

$$\forall x \in I(|x - x_0| < \delta \implies |f(x) - f(x_0)| < \epsilon)$$

The function is called continuous if it is continuous at every point in its domain.

Rem. 102. The **intuition** behind this definition is that for any given distance $\epsilon > 0$ around $f(x_0)$, we can find a $\delta > 0$ such that for all $x \in D$ that are at least δ close to x_0 , we have $|f(x) - f(x_0)| < \epsilon$. So for any bound on the output (ϵ) there exists a bound on the input (δ) such that all elements in the input bound $(x_0 - \delta, x_0 + \delta)$ are mapped to the output bound $(f(x_0) - \epsilon, f(x_0) + \epsilon)$.

Def. 103. Let $f : D \rightarrow \mathbb{R}$ be a function. Then f is called **left(right)-continuous** at x_0 if

$$\lim_{x \rightarrow x_0^{-(+)}} f(x) = f(x_0)$$

Thm. 104. Let $f : D \subset \mathbb{D}(f) \subset \mathbb{R}$ be a function and $x_0 \in D$. f is continuous at x_0 iff for every sequence $(x_n)_{n \in \mathbb{N}} \subset D$ with $\lim_{n \rightarrow \infty} x_n = x_0$ it holds that $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$.

Rem. 105. Continuity can also be viewed in terms of intervals, where a function f is continuous iff for every open interval $I \subset \mathbb{R}$ it holds that $f^{-1}(I)$ is open.

Lem. 106. Let f, g be continuous functions then

$$f + g, \quad f - g, \quad f \cdot g, \quad c \cdot f, \quad \frac{f}{g}, \quad f \circ g$$

are continuous

Thm. 107. Let I be an interval and let $f : I \rightarrow J$ be a continuous and bijective function. Then J is an interval too and the inverse function $f^{-1} : J \rightarrow I$ is continuous.

Thm. 108. Let $f : D \rightarrow \mathbb{R}$ be a continuous function. Let $c \in [f(a), f(b)]$ then there exists a $x_0 \in [a, b]$ such that $f(x_0) = c$.

Lem. 109. Let $[a, b]$ be a compact interval and let $(x_n)_{n \in \mathbb{N}} \subset [a, b]$ be a sequence. Then there exists a subsequence $(x_{n_k})_{k \in \mathbb{N}}$ such that

$$\lim_{k \rightarrow \infty} x_{n_k} = x_0 \quad \text{for some } x_0 \in [a, b].$$

Def. 110. Let $f : D \subset \mathbb{R} \rightarrow \mathbb{R}$ be a function. If for $x_0 \in D$ it holds that $f(x_0) \geq f(x) \forall x \in D$, then f has a local maximum at x_0 . If for $x_0 \in D$ it holds that $f(x_0) \leq f(x) \forall x \in D$, then f has a local minimum at x_0 .

Thm. 111. Let $f : I \subset \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function and I be a compact interval. Then f is bounded and attains its maximum and minimum value on I . Hence

$$\exists x_{\min}, x_{\max} \in I \text{ such that}$$

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}) \quad \forall x \in I$$

4. Differential calculus

Def. 112. The **average rate of change** of a function f in the domain between a and $a + h$:=

$$\frac{\Delta y}{\Delta x} = \frac{f(a + h) - f(a)}{h}$$

and also called the **difference quotient**.

Def. 113. The **derivative** f' of a function f at the coordinate a :=

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} &= \lim_{b \rightarrow a} \frac{f(b) - f(a)}{b - a} \\ &= \frac{df}{dx} \Big|_{x=a} = f'(a) \end{aligned}$$

Rem. 114. The derivative $f'(a)$ can geometrically interpreted as the incline/decline of the graph of f at the point $(a, f(a))$.

Def. 115. If for a function f at the coordinate a the derivative $f'(a)$ exists then f is called **differentiable** at a . If f is differentiable for every $a \in \mathbb{D}$ then f is called differentiable.

Lem. 116. If f is differentiable at x_0 then f is continuous at x_0 .

Def. 117. If the limit

$$\lim_{h \rightarrow 0^+} \frac{f(a + h) - f(a)}{h}$$

exists then we call it the **right-sided derivative** (or the left-sided derivative analogue)

Def. 118. For a function f we can iteratively define higher derivatives for $n \in \mathbb{N}_0$

$$f^{(1)} = f', f^{(2)} = f'', \dots, f^{(n+1)} = (f^{(n)})'$$

if $f^{(n)}$ exists we call it **n -times differentiable**.

Def. 119. if $f^{(n)}$ is continuous we call f **n -times continuous differentiable**. The set of all n -times continuous differentiable functions on D is denoted as $C^n(D)$

Def. 120. If for a function f and some $x_0 \in \mathbb{D}$ and $\delta > 0$ we have

$$f(x_0) \geq (\leq) f(x) \quad \forall x \in \mathbb{D} \cap (x_0 - \delta, x_0 + \delta)$$

then x_0 is called a **local maximum (minimum)**.

4.1 Derivation rules

Thm. 121. For functions $f(x)$ we have

$$f(x) = ax^p \implies f'(x) = apx^{p-1}$$

$$f(x) = a^x \implies f'(x) = \ln(a) \cdot a^x$$

$$f(x) = \sin(x) \implies f'(x) = \cos(x)$$

$$f(x) = \cos(x) \implies f'(x) = -\sin(x)$$

$$f(x) = \tan(x) \implies f'(x) = \frac{1}{\cos^2(x)}$$

$$f(x) = \sinh(x) \implies f'(x) = \cosh(x)$$

$$f(x) = \cosh(x) \implies f'(x) = \sinh(x)$$

$$f(x) = \ln(x) \implies f'(x) = \frac{1}{x}$$

$$f(x) = \arcsin(x) \implies f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$f(x) = \arccos(x) \implies f'(x) = -\frac{1}{\sqrt{1-x^2}}$$

$$f(x) = \arctan(x) \implies f'(x) = \frac{1}{1+x^2}$$

Thm. 122. Sum Rule: For the n -th derivative of a sum:

$$(f + g)^{(n)} = f^{(n)} + g^{(n)}$$

Thm. 123. General Leibniz Rule (Product): The n -th derivative of a product is given by:

$$(f \cdot g)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(n-k)} g^{(k)}$$

Thm. 124. Chain Rule: For the composition of functions:

$$\begin{aligned} (f \circ g)'(x) \\ = f'(g(x)) \cdot g'(x) \end{aligned}$$

Thm. 125. Quotient Rule: For $g(x) \neq 0$:

$$\begin{aligned} \left(\frac{f}{g}\right)' \\ = \frac{f'g - fg'}{g^2} \end{aligned}$$

Thm. 126. Inverse Functions: If f is differentiable and injective:

$$\begin{aligned} (f^{-1})'(x) \\ = \frac{1}{f'(f^{-1}(x))} \end{aligned}$$

4.2 Bernoulli-L'Hopital

Thm. 127. Let $I \subset \mathbb{R}$ be an interval, $f : I \rightarrow \mathbb{R}$, and let x_0 be a local extremum of f . Then one of the following holds:

1. x_0 is an endpoint of I
2. f is not differentiable at x_0
3. f is differentiable at x_0 and $f'(x_0) = 0$

Thm. 128. Rolle's Theorem: Let $a < b$ with $f(a) = f(b)$, and let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then $\exists \xi \in (a, b)$ with

$$f'(\xi) = 0$$

Thm. 129. Mean-Value-Theorem: Let $a < b$, and let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then $\exists \xi \in (a, b)$ with

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}$$

Thm. 130. Cauchy's Mean-Value-Theorem: Let $a < b$, and let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then $\exists \xi \in (a, b)$ with

$$\begin{aligned} \frac{f(b) - f(a)}{g(b) - g(a)} \\ = \frac{f'(\xi)}{g'(\xi)} \end{aligned}$$

Thm. 131. Bernoulli-L'Hopital's Rule: Let $f, g : I \rightarrow \mathbb{R}$ be differentiable functions on (a, b) and let $x_0 \in (a, b)$. If $g(x) \neq 0$ and $g'(x) \neq 0$ for all $x \in (a, b)$. If

$$\lim_{x \rightarrow x_0} |f(x)| = \lim_{x \rightarrow x_0} |g(x)| \in \{0, \infty\}$$

$$\text{and } \exists L = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)}$$

then

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = L$$

4.3 Convex and Concave graphs

Def. 132. A function $f : I \rightarrow \mathbb{R}$ is called **convex** iff $\forall a, b \in I \forall t \in (0, 1)$

$$f((1-t)a + bt) \leq (1-t)f(a) + tf(b)$$

Def. 133. A function $f : I \rightarrow \mathbb{R}$ is called **concave** iff $\forall a, b \in I \forall t \in (0, 1)$

$$f((1-t)a + bt) \geq (1-t)f(a) + tf(b)$$

Thm. 134. Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \rightarrow \mathbb{R}$ be differentiable. Then $\forall x$

$$f'(x) \geq 0 \iff f \text{ is increasing}$$

Thm. 135. Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \rightarrow \mathbb{R}$ be differentiable. Then

$$f \text{ is convex} \iff f' \text{ is increasing}$$

Thm. 136. Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \rightarrow \mathbb{R}$ be twice differentiable. Then

$$f \text{ is convex} \iff f'' \geq 0$$

4.4 Taylor Series

Lemma. 137. We can locally approximate the graph of f at x_0 with the tangent line $t(x)$:

$$\begin{aligned} f(x) &\approx t(x) \\ &= f(x_0) + f'(x_0)(x - x_0) \end{aligned}$$

Lemma. 138. To even more accurately approximate the function f we can add further terms to the tangent line expansion:

$$\begin{aligned} f(x) &\approx t(x) \\ &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 \end{aligned}$$

Thm. 139. Taylor's Theorem: Let $f : (a, b) \rightarrow \mathbb{R}$ be $n+1$ times differentiable. Then

$$f(x) = P_n(x) + R(x)$$

with $P_n(x)$ the Taylor polynomial of degree n and $R(x)$ the remainder term:

$$\begin{aligned} &P_n(x) \\ &= \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \\ &R(x) \\ &= \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1} \quad \text{for some } \xi \in (x_0, x) \text{ or } (x, x_0) \end{aligned}$$

Def. 140. The **Taylor series expansion** of an infinitely differentiable function $f : (a, b) \rightarrow \mathbb{R}$ at x_0 is given by:

$$\begin{aligned} &f(x) \\ &= \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \end{aligned}$$

Thm. 141. Standard Taylor Series: The following converge for all $x \in \mathbb{R}$:

$$\begin{aligned} e^x &= \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \\ \sin x &= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \\ \cos x &= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \end{aligned}$$

Imp. 142. Not all infinitely differentiable functions have a Taylor series expansion that converges to the function itself. For example, $\ln(1+x)$, $\frac{1}{1-x}$, and $(1+x)^p$ only converge for $-1 < x < 1$.

Thm. 143. Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ and $g(x) = \sum_{n=0}^{\infty} b_n x^n$ be two power series with radius of convergence R . Then:

$$\begin{aligned} &f(x)g(x) \\ &= \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) x^n \end{aligned}$$

Thm. 144. Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence R . Then:

$$\begin{aligned} &f'(x) \\ &= \sum_{n=1}^{\infty} n a_n x^{n-1} \end{aligned}$$

Thm. 145. Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence R . Then:

$$\begin{aligned} &\int f(x) dx \\ &= \sum_{n=0}^{\infty} \int a_n x^n dx \end{aligned}$$

5. Integration

5.1 Indefinite Integrals and Techniques

Def. 146. Functions $F : I \rightarrow \mathbb{R}$ such that $F'(x) = f(x)$ for all $x \in (a, b)$ are called **antiderivatives** of f .

Def. 147. The family of all anti-derivatives of a function $f : I \rightarrow \mathbb{R}$ for $I \subseteq \mathbb{R}$ an open interval is called the **indefinite integral** and $:=$

$$\begin{aligned} &\int f(x) dx \\ &= \{F : I \rightarrow \mathbb{R} : F'(x) = f(x)\} = F(x) + C \end{aligned}$$

Thm. 148. Let $f, g : I \rightarrow \mathbb{R}$ be integrable functions and let $c \in \mathbb{R}$. Then:

$$\begin{aligned} &\int (f(x) \pm g(x)) dx \\ &= \int f(x) dx \pm \int g(x) dx \\ &\int c f(x) dx = c \int f(x) dx \end{aligned}$$

Thm. 149. Basic Antiderivatives:

$$\begin{aligned} \int x^p dx &= \frac{x^{p+1}}{p+1} + C \quad (p \neq -1) \\ \int \frac{1}{x} dx &= \ln|x| + C \\ \int e^x dx &= e^x + C \\ \int a^x dx &= \frac{a^x}{\ln(a)} + C \\ \int \sin(x) dx &= -\cos(x) + C \\ \int \cos(x) dx &= \sin(x) + C \\ \int \frac{1}{1+x^2} dx &= \arctan(x) + C \end{aligned}$$

Thm. 150. Integration by Parts: For differentiable functions $f(x)$ and $g(x)$ we have:

$$\begin{aligned} & \int f'(x)g(x) dx \\ &= f(x)g(x) - \int f(x)g'(x) dx \end{aligned}$$

Thm. 151. Substitution Rule: For functions $f(x)$ and $g(x)$ we have:

$$\begin{aligned} & \int f(g(x))g'(x) dx \\ &= \int f(u) du = F(g(x)) + C \end{aligned}$$

where F is an antiderivative of f and $u = g(x)$.

5.2 The Definite Integral (Riemann Integral)

Def. 152. Let U and L denote the upper and lower sums of a bounded function f . Then f is called **Riemann-integrable** iff $\inf U = \sup L$. The value of the definite integral $\int_a^b f(x) dx$ is then this common value.

Thm. 153. Integrability Conditions: Any continuous or monotone function on $[a, b]$ is Riemann-integrable.

Thm. 154. Properties of Riemann Integral: Let f, g be integrable and $c \in \mathbb{R}$:

$$\begin{aligned} \int_a^b (f(x) \pm g(x)) dx &= \int_a^b f(x) dx \pm \int_a^b g(x) dx \\ \int_a^b cf(x) dx &= c \int_a^b f(x) dx \\ f(x) \leq g(x) &\implies \int_a^b f(x) dx \leq \int_a^b g(x) dx \\ \left| \int_a^b f(x) dx \right| &\leq \int_a^b |f(x)| dx \\ \int_a^b f(x) dx &= - \int_b^a f(x) dx \\ \int_a^c f(x) dx &= \int_a^b f(x) dx + \int_b^c f(x) dx \end{aligned}$$

Thm. 155. Mean Value Theorem for Integrals: Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then there exists a $c \in [a, b]$ such that:

$$\int_a^b f(x) dx = (b - a)f(c)$$

Thm. 156. Fundamental Theorem of Integral Calculus: Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then for all $C \in \mathbb{R}$, $F(x)$ is an antiderivative of f where:

$$F(x) = \int_a^x f(y) dy + C$$

5.3 Sequences of Functions and Integrals

Def. 157. Let $(f_n)_{n \in \mathbb{N}_0}$ be a sequence of functions. **Pointwise Convergence:** It converges pointwise to f if for every x , the sequence of real numbers $(f_n(x))$ converges to $f(x)$.

Def. 158. Uniform Convergence: It converges uniformly to f if for every $\epsilon > 0$, there exists an N such that for all $n \geq N$ and **for all** x simultaneously, $|f_n(x) - f(x)| < \epsilon$.

Thm. 159. Swapping Limit and Integral: If a sequence of integrable functions (f_n) converges **uniformly** to f on $[a, b]$, then the limit function f is also integrable, and:

$$\begin{aligned} & \int_a^b f(x) dx \\ &= \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx \end{aligned}$$

5.4 Definite Integration Techniques

Lem. 160. If $F : [a, b] \rightarrow \mathbb{R}$ is continuously differentiable, then:

$$\begin{aligned} F(x) &= F(a) + \int_a^x F'(t) dt \\ \int_a^b F'(t) dt &= F(b) - F(a) = \left[F(t) \right]_a^b \end{aligned}$$

Lem. 161. Substitution for Definite Integrals: Let f be continuous and g continuously differentiable.

$$\begin{aligned} & \int_a^b f(g(x))g'(x) dx \\ &= \int_{g(a)}^{g(b)} f(u) du \end{aligned}$$

Version 2: If $f'(x) \neq 0$ and f^{-1} is the inverse of f :

$$\begin{aligned} & \int_a^b g(f(x)) dx \\ &= \int_{f(a)}^{f(b)} g(y)(f^{-1})'(y) dy \end{aligned}$$

Lem. 162. Integration by Parts for Definite Integrals: Let f, g be continuously differentiable.

$$\begin{aligned} & \int_a^b f'(x)g(x) dx \\ &= \left[f(x)g(x) \right]_a^b - \int_a^b f(x)g'(x) dx \end{aligned}$$

6. Differential Equations

6.1 Modelling and Definitions

Rem. 163. Modelling Process: To derive an ODE from a physical scenario:

1. Identify the quantity $u(t)$ and express its change over a small interval Δt : $u(t + \Delta t) = u(t) + (\text{rate}) \cdot \Delta t$
2. Rearrange and take the limit $\Delta t \rightarrow 0$: $\frac{u(t+\Delta t) - u(t)}{\Delta t} = \text{rate} \implies u'(t) = \text{rate}$

Rem. 164. Typical ODE Models:

1. Bounded Growth (Logistic): With capacity L :

$$u'(t) = ku(t) \left(1 - \frac{u(t)}{L} \right)$$

2. Bimolecular Reaction ($A + B \rightarrow C$): With concentrations a, b :

$$u'(t) = k(a - u(t))(b - u(t))$$

Def. 165. An ordinary differential equation (ODE) contains derivatives of a function of one independent variable. It is:

- **of n -th order:** if the n -th derivative is the highest occurring.
- **linear:** if the unknown function $u(t)$ and its derivatives appear only as a linear combination.
- **homogeneous:** if it only contains terms with $u(t)$ and its derivatives (otherwise **inhomogeneous**).

Thm. 166. Picard-Lindelöf (Existence and Uniqueness): If $f(x, y)$ is continuous and Lipschitz continuous with respect to y , then the IVP $y' = f(x, y)$ with $y(x_0) = y_0$ has a unique solution in some interval around x_0 .

6.2 First-Order ODEs

Thm. 167. For an ODE of the form $y' = f(x)g(y)$, we can separate variables:

$$\begin{aligned} & \frac{dy}{dx} = f(x)g(y) \\ \implies \int \frac{1}{g(y)} dy &= \int f(x) dx + C \end{aligned}$$

Thm. 168. If $y' = f(ax + by + c)$, substitute $u = ax + by + c$. Since $u' = a + by'$, this yields the separable ODE:

$$u' = a + bf(u)$$

Thm. 169. If $y' = g\left(\frac{y}{x}\right)$, substitute $u = \frac{y}{x}$. Since $y' = u'x + u$, this yields the separable ODE:

$$u'x = g(u) - u$$

6.3 Linear ODEs of n -th Order

Thm. 170. Superposition Principle: If y_1 and y_2 solve a linear homogeneous ODE, then for any $C_1, C_2 \in \mathbb{R}$:

$$y(x) = C_1y_1(x) + C_2y_2(x)$$

Thm. 171. An n -th order linear homogeneous ODE has n linearly independent solutions $y_1, \dots, y_n$. Its general solution is:

$$y(x) = \sum_{i=1}^n C_i y_i(x) \quad \text{with } C_i \in \mathbb{R}$$

Lem. 172. For a homogeneous ODE with constant coefficients $a_n y^{(n)} + \dots + a_0 y = 0$, the Ansatz $y(x) = e^{\lambda x}$ yields the characteristic equation:

$$a_n \lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_1 \lambda + a_0 = 0$$

with characteristic polynomial $p(\lambda)$.

Thm. 173. General Solution from Characteristic Roots: The roots of $p(\lambda)$ determine the fundamental solutions:

- **Distinct real root** $\lambda \in \mathbb{R}$

$$y(x) = e^{\lambda x}$$

- **Multiple real root** $\lambda \in \mathbb{R}$ with multiplicity m

$$e^{\lambda x}, x e^{\lambda x}, \dots, x^{m-1} e^{\lambda x}$$

- **Complex pair** $\lambda = \alpha \pm i\beta \in \mathbb{C}$ (multiplicity m)

$$e^{\alpha x} \cos(\beta x), e^{\alpha x} \sin(\beta x), \dots,$$

$$x^{m-1} e^{\alpha x} \cos(\beta x), x^{m-1} e^{\alpha x} \sin(\beta x)$$

6.4 Inhomogeneous Linear ODEs

Lem. 174. The general solution of $a_n y^{(n)} + \dots + a_0 y = s(x)$ is $y(x) = y_h(x) + y_p(x)$, where y_h is the general homogeneous solution and y_p is a particular solution.

6.4.1 Method of Undetermined Coefficients (Ansatz)

Thm. 175. Method of Undetermined Coefficients: For right-hand side $s(x)$, choose the particular Ansatz $y_p(x)$ (without resonance):

- **Polynomial** $P_m(x)$:

$$y_p(x) = C_m x^m + \dots + C_1 x + C_0$$

- **Exponential** $A e^{kx}$:

$$y_p(x) = C e^{kx}$$

- **Trig** $A \sin(\omega x) + B \cos(\omega x)$:

$$y_p(x) = C_1 \sin(\omega x) + C_2 \cos(\omega x)$$

Imp. 176. Resonance: If the term e^{kx} or $e^{i\omega x}$ in $s(x)$ matches an m -fold root of the characteristic polynomial $p(\lambda)$, multiply the standard Ansatz $y_p(x)$ by x^m .

6.4.2 Variation of Constants

Thm. 177. Variation of Constants (First-Order Linear ODE): For $y' + p(x)y = q(x)$, the general solution is:

$$y(x) = e^{-P(x)} \left(\int q(x) e^{P(x)} dx + C \right)$$

where $P(x) = \int p(x) dx$ is an antiderivative of $p(x)$.

6.5 Initial Value Problems and Boundary-Value Problems

Def. 178. An **Initial Value Problem (IVP)** specifies conditions at a single point x_0 (e.g. $y(x_0) = y_0, y'(x_0) = y_1$). A **Boundary Value Problem (BVP)** specifies conditions at multiple points (e.g. $y(a) = y_a, y(b) = y_b$).